```
#include <iostream>
#include <string>
using namespace std;

struct Patient
{
    int id;
    string name;
    int age;
    string disease;
    Patient *next;
};

Patient *head = NULL;

void insertPatient()
{
    Patient *newNode = new Patient;

    cout << "Enter Patient ID: ";
    cin >> newNode->id;

    cout << "Enter Patient Name: ";
    cin >> newNode->name;

    cout << "Enter Age: ";
    cin >> newNode->age;

    cout << "Enter Disease: ";
    cin >> newNode->disease;

    newNode->next = NULL;

    if (head == NULL)
        head = newNode;
    else
    {
        Patient *temp = head;

        while (temp->next != NULL)
            temp = temp->next;

        temp->next = newNode;
    }

    cout << "Patient record inserted successfully.\n";
}
```

```
void displayPatients()
{
    if (head == NULL)
    {
        cout << "No patient records found.\n";
        return;
    }

    Patient *temp = head;

    cout << "\nPatient Records:\n";

    while (temp != NULL)
    {
        cout << "ID: " << temp->id << endl;
        cout << "Name: " << temp->name << endl;
        cout << "Age: " << temp->age << endl;
        cout << "Disease: " << temp->disease << endl;
        cout << "------------------\n";

        temp = temp->next;
    }
}

void searchPatient()
{
    int id;

    cout << "Enter Patient ID to search: ";
    cin >> id;

    Patient *temp = head;

    while (temp != NULL)
    {
        if (temp->id == id)
        {
            cout << "Patient Found!\n";
            cout << "Name: " << temp->name << endl;
            cout << "Age: " << temp->age << endl;
            cout << "Disease: " << temp->disease << endl;
            return;
        }

        temp = temp->next;
    }

    cout << "Patient not found.\n";
```

```
}

void deletePatient()
{
    int id;

    cout << "Enter Patient ID to delete: ";
    cin >> id;

    Patient *temp = head;
    Patient *prev = NULL;

    while (temp != NULL && temp->id != id)
    {
        prev = temp;
        temp = temp->next;
    }

    if (temp == NULL)
    {
        cout << "Patient not found.\n";
        return;
    }

    if (prev == NULL)
        head = temp->next;
    else
        prev->next = temp->next;

    delete temp;

    cout << "Patient record deleted successfully.\n";
}

int main()
{
    int choice;

    do
    {
        cout << "\n--- Hospital Patient Record Management ---\n";
        cout << "1. Insert Patient\n";
        cout << "2. Display Patients\n";
        cout << "3. Search Patient\n";
        cout << "4. Delete Patient\n";
        cout << "5. Exit\n";

        cout << "Enter your choice: ";
```

```
        cin >> choice;

        switch (choice)
        {
        case 1:
            insertPatient();
            break;

        case 2:
            displayPatients();
            break;

        case 3:
            searchPatient();
            break;

        case 4:
            deletePatient();
            break;

        case 5:
            cout << "Program terminated.\n";
            break;

        default:
            cout << "Invalid choice.\n";
        }

    } while (choice != 5);

    return 0;
}
```

--- Hospital Patient Record Management ---
1. Insert Patient
2. Display Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter your choice: 1

Enter Patient ID: 101
Enter Patient Name: Rahul
Enter Age: 25
Enter Disease: Fever
Patient record inserted successfully.

--- Hospital Patient Record Management ---
1. Insert Patient
2. Display Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter your choice: 1

Enter Patient ID: 102
Enter Patient Name: Amit
Enter Age: 30
Enter Disease: Cold
Patient record inserted successfully.

--- Hospital Patient Record Management ---
1. Insert Patient
2. Display Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter your choice: 2

Patient Records:
ID: 101
Name: Rahul
Age: 25
Disease: Fever

ID: 102
Name: Amit
Age: 30
Disease: Cold

--- Hospital Patient Record Management ---
1. Insert Patient
2. Display Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter your choice: 3

Enter Patient ID to search: 102
Patient Found!
Name: Amit
Age: 30
Disease: Cold

--- Hospital Patient Record Management ---
1. Insert Patient
2. Display Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter your choice: 4

Enter Patient ID to delete: 101
Patient record deleted successfully.

--- Hospital Patient Record Management ---
1. Insert Patient
2. Display Patients
3. Search Patient
4. Delete Patient
5. Exit
Enter your choice: 5

Program terminated.

Algorithm: Hospital Patient Record Management

1.Start

2.Initialize head = NULL.

3.Display the menu:
•Insert Patient
•Display Patients
•Search Patient
•Delete Patient
•Exit

4. Read the user's choice.

5. If choice is 1 (Insert):
•Create a new patient node.
•Enter Patient ID, Name, Age and Disease.
•Add the node at the end of the linked list.
•Display “Patient record inserted successfully.”

6. If choice is 2 (Display):
•Check whether the list is empty.
•If not empty, traverse the list and display all patient records.

7. If choice is 3 (Search):
•Enter Patient ID.
•Traverse the list and compare each ID.
•If found, display patient details.
•Otherwise display “Patient not found.”

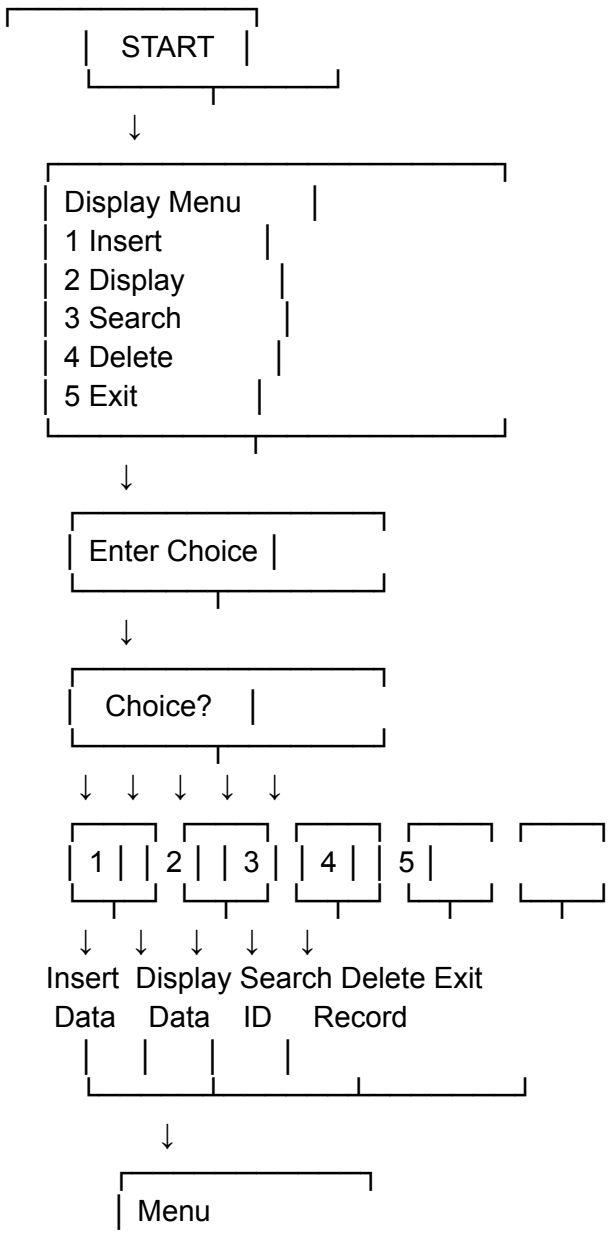
8. If choice is 4 (Delete):
•Enter Patient ID.
•Search for the patient node.
•If found, remove the node and delete it.
•Otherwise display “Patient not found.”

9. If choice is 5, display “Program terminated.”

10. Repeat the menu until choice 5 is selected.

11. Stop.

Flowchart :


START
Display Menu
1 Insert
2 Display
3 Search
4 Delete
5 Exit
Enter Choice
Choice?
1
2
3
4
5
Insert Data
Display Data
Search ID
Delete Record
Exit
Menu